Datasets Reference

This document lists all datasets used or planned for the Multi-Agent Medical Assistant project — existing, in-progress, and future extensions. Organized by agent/disease.

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Already Implemented

1. COVID-19 Chest X-ray

Field	Details
Agent	`CHEST_XRAY_AGENT`
Model	`agents/image_analysis_agent/chest_xray_agent/models/covid_chest_xray_model.pth`
Status	Trained and integrated

What it contains: Chest X-ray images labeled as COVID-19 positive or Normal. Binary classification dataset widely used for COVID-19 radiological screening research.

Classes: `covid19` / `normal`

Task: Binary image classification

How it's used: The trained `.pth` model (PyTorch) is loaded at inference time. The `CHEST_XRAY_AGENT` runs the uploaded image through the model and returns a positive/negative COVID-19 result, followed by human validation.

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2. Skin Lesion (ISIC)

Field	Details
Agent	`SKIN_LESION_AGENT`
Model	`agents/image_analysis_agent/skin_lesion_agent/models/checkpointN25_.pth.tar`
Status	Trained and integrated

What it contains: Dermoscopic images of skin lesions from the ISIC (International Skin Imaging Collaboration) archive. The model performs semantic segmentation — producing a binary mask that highlights the lesion region.

Task: Semantic segmentation

How it's used: The model segments the lesion area from the background and outputs a segmentation plot saved to `uploads/skin_lesion_output/segmentation_plot.png`, which is returned to the frontend alongside the text response. Human validation is required.

Placeholder — Needs Dataset & Model

3. Brain Tumor MRI

Field	Details
Agent	`BRAIN_TUMOR_AGENT`
Model	Not yet implemented (stub in `agents/agent_decision.py:455`)
Status	Agent routing exists; model returns a placeholder string

Recommended Dataset

Brain Tumor MRI Dataset — Kaggle (Masoud Nickparvar)

What it contains: 7,023 MRI brain scan images across 4 classes collected from multiple sources (figshare, SARTAJ, Br35H). Standard JPEG format, pre-organized into train/test folders.

Class	Description
`glioma`	Malignant tumor in glial cells
`meningioma`	Tumor in meninges (usually benign)
`pituitary`	Tumor in the pituitary gland
`no_tumor`	Healthy brain MRI

Total images: ~7,023
Format: JPEG, variable resolution (resized to 224×224 for training)
Access: Free, no registration required
License: Public domain / research use

Task: Multi-class classification (4 classes)

How it will be used: Train an EfficientNet/ResNet classifier on the 4 classes. Replace the current placeholder response in `run_brain_tumor_agent()` with actual model inference. Human validation appended as with other CV agents.

Alternative — Segmentation approach: For tumor localization use **BraTS Challenge** (Brain Tumor Segmentation) — provides ground truth segmentation masks but requires registration.

RAG Knowledge Base (PDF Documents)

These are ingested into the Qdrant vector store via `ingest_rag_data.py` and used by the `RAG_AGENT`.

Document	Location	Covers
`brain_tumor_2024.pdf`	`data/raw/`	Deep learning techniques for brain tumor detection
`brain_tumors_ucni.pdf`	`data/raw/`	Introduction to brain tumors
`covid_chest_xray_2024.pdf`	`data/raw/`	Deep learning for COVID-19 detection from CXR
`skin_lesion_2023.pdf`	`data/raw/`	Skin lesion classification techniques

To extend the RAG knowledge base: Drop any medical PDF into `data/raw/` and run:

```
python ingest_rag_data.py --file ./data/raw/<filename>.pdf
```

Planned Extensions — Image Agents

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4. Tuberculosis — Chest X-ray

Primary Dataset

NIH Chest X-ray Dataset — Kaggle

What it contains: 112,120 frontal-view chest X-rays from 30,805 unique patients. Each image has up to 14 disease labels extracted from radiology reports using NLP. TB is included as one of the labels.

Detail	Value
Total images	112,120
Format	PNG, 1024×1024
Size	~42 GB
Access	Free on Kaggle
License	Public domain (NIH)

Classes used for TB agent: `Infiltration` / `No Finding` (filter TB-relevant labels)

Supplementary Datasets

Shenzhen & Montgomery TB Chest X-ray — Kaggle

Smaller dedicated TB datasets (662 + 138 images) with binary labels (TB / Normal). Useful for fine-tuning.

Task: Binary classification (TB positive / negative)

How it will be used: Train DenseNet121/EfficientNet on TB vs. Normal labels. New agent `tb_agent/` following the same pattern as `chest_xray_agent/`. Reuses existing CXR preprocessing pipeline.

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5. Pneumothorax — Chest X-ray

SIIM-ACR Pneumothorax Segmentation — Kaggle

What it contains: 12,954 chest X-ray images with pixel-level segmentation masks for pneumothorax (collapsed lung) regions. Part of the SIIM-ACR 2019 Kaggle competition.

Detail	Value
Total images	12,954
Format	DICOM → PNG (after conversion)
Masks	RLE-encoded binary masks
Access	Free on Kaggle (account required)
License	Research use

Task: Segmentation (pneumothorax region) or binary presence classification

How it will be used: For classification: train EfficientNet on presence/absence. For segmentation: train UNet on pixel masks. New agent `pneumothorax_agent/` — can share an agent folder with TB as `chest_xray_extended_agent/`.

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6. Multi-label Chest X-ray Findings

CheXpert Dataset — Stanford ML Group

What it contains: 224,316 chest X-rays from 65,240 patients, labeled for 14 pathologies using an automated labeler. One of the largest CXR datasets available.

Detail	Value
Total images	224,316
Format	JPEG
Size	~439 GB (full) / ~11 GB (small version)
Access	Free — requires registration (instant)

License	Research use only

Classes (14 labels): Cardiomegaly, Edema, Consolidation, Atelectasis, Pleural Effusion, and 9 others

Uncertainty labels: `positive` / `negative` / `uncertain` per finding

Task: Multi-label classification (multiple conditions per image)

How it will be used: Train a multi-label EfficientNet (sigmoid output, one node per class). The agent returns a structured list of detected findings with confidence scores per finding. Use the **small version** (~11GB) for faster iteration.

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7. Diabetic Retinopathy

APTOS 2019 Blindness Detection — Kaggle

What it contains: 3,662 retinal fundus photographs taken with fundus cameras under various imaging conditions in India. Graded by clinicians on a 0–4 severity scale.

Detail	Value
Total images	3,662
Format	PNG, variable resolution
Size	~9 GB
Access	Free on Kaggle (account required)
License	Competition terms (research use)

Classes (DR grading scale):

Grade	Severity
0	No DR
1	Mild
2	Moderate
3	Severe
4	Proliferative DR

Task: Ordinal classification (5-class grading)

Supplementary: **EyePACS — Kaggle** — 88,702 images, same grading scale, much larger but noisier labels.

How it will be used: Train EfficientNet-B4 with preprocessing: CLAHE contrast enhancement + circular crop to remove black borders. Output: DR grade + severity description. New agent `diabetic_retinopathy_agent/`. Uses fundus image type routing (update `image_classifier.py` prompt to recognize fundus images).

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8. Mammography — Breast Cancer

CBIS-DDSM Breast Cancer Image Dataset — Kaggle

What it contains: Preprocessed version of the Curated Breast Imaging Subset of DDSM (Digital Database for Screening Mammography). Converted from DICOM to PNG, organized into train/test splits. Two sub-tasks: mass detection and calcification detection.

Detail	Value
Total images	~10,239 (ROI crops + full mammograms)
Format	PNG (pre-converted from DICOM)
Size	~6.8 GB
Access	Free on Kaggle — no DUA required
License	Public research use

Sub-datasets & Classes:

Sub-task	Classes
Mass classification	`BENIGN` / `MALIGNANT`
Calcification classification	`BENIGN` / `MALIGNANT`
Mass detection	Bounding box coordinates

ROI crops are pre-extracted — no tiling pipeline needed.

Task: Binary classification (benign vs. malignant) per finding type

How it will be used: Train two separate EfficientNet classifiers — one for masses, one for calcifications. New agent `mammography_agent/` with a two-stage output: finding type (mass/calcification) + malignancy prediction. Human validation mandatory (clinical sensitivity). Update `image_classifier.py` to recognize mammography images.

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9. Histopathology — Colorectal Cancer Tissue

NCT-CRC-HE-100K — Zenodo (Kather et al.)

What it contains: 100,000 non-overlapping image patches from hematoxylin & eosin (H&E) stained histological images of human colorectal cancer and normal tissue. All patches are 224×224 pixels at 0.5 microns/pixel.

Detail	Value
Total images	100,000
Format	PNG, 224×224 pixels
Size	~7.4 GB
Access	Free on Zenodo (no registration)
License	CC BY 4.0

Classes (9 tissue types):

Label	Tissue
ADI	Adipose
BACK	Background
DEB	Debris
LYM	Lymphocytes
MUC	Mucus
MUS	Smooth muscle
NORM	Normal colon mucosa
STR	Cancer-associated stroma
TUM	Colorectal adenocarcinoma epithelium

Task: Multi-class patch classification (9 classes)

Supplementary — Nuclei Segmentation: **PanNuke — from the Histopathology Datasets Index** 189,744 nuclei patches across 19 tissue types. For nuclei-level analysis if needed.

How it will be used: Train EfficientNet on 9-class tissue classification (patch-level). Input: cropped histopathology image patch. Output: tissue type distribution and tumor presence confidence. New agent `histopathology_agent/`. Standard JPEG/PNG input — no OpenSlide or WSI pipeline needed (patch-based only). Update `image_classifier.py` to recognize histopathology slide images.

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10. Musculoskeletal Fracture Detection

MURA — Stanford ML Group

What it contains: 40,561 musculoskeletal X-ray images from 14,863 studies across 7 body parts. Each study is labeled as normal or abnormal (fracture/lesion) by board-certified radiologists.

Detail	Value
Total images	40,561
Format	PNG
Size	~6 GB
Access	Free — requires registration (instant)
License	Research use only

Body parts covered: Elbow, Finger, Forearm, Hand, Humerus, Shoulder, Wrist

Classes: `normal` / `abnormal` (per body part)

Task: Binary classification per body part study

How it will be used: Train DenseNet169 (architecture from original MURA paper) per body part, or a single model with body part as input context. New agent `fracture_agent/`. Output: abnormality detected / not detected + body part identified. Update `image_classifier.py` to recognize orthopedic X-ray images.

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11. Age-related Macular Degeneration — OCT

Retinal OCT Images — Kermany 2018 — Kaggle

What it contains: 84,495 validated retinal OCT images organized into train/test folders. Collected from multiple hospitals, labeled by medical experts with a 2-stage quality check.

Detail	Value
Total images	84,495
Format	JPEG
Size	~6.3 GB
Access	Free on Kaggle
License	CC BY 4.0

Classes (4 retinal conditions):

Class	Description
`CNV`	Choroidal neovascularization (wet AMD)
`DME`	Diabetic macular edema
`DRUSEN`	Early/intermediate dry AMD (drusen deposits)
`NORMAL`	Healthy retina

Task: Multi-class classification (4 classes)

> **Note:** The Farsiu 2014 dataset (also by the same uploader) was evaluated but rejected — it stores data in MATLAB `.mat` format and focuses on retinal layer boundary segmentation rather than disease classification, making it a poor fit for this project.

How it will be used: Train EfficientNet-B3 on 4-class OCT classification. New agent `oct_agent/` (or `amd_agent/`). Output: condition name + clinical description. Update `image_classifier.py` to recognize retinal OCT scan images.

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12. Diabetic Foot Ulcer

Primary Dataset

Diabetic Foot Ulcer (DFU) — Kaggle (laithjj)

What it contains: ~1,426 high-definition images of diabetic foot ulcers collected from DFUC 2020 and Xiangya Hospital. Labeled across 4 clinical categories.

Detail	Value

Total images	~1,426
Format	JPEG/PNG
Size	~500 MB
Access	Free on Kaggle
License	CC BY-NC

Classes: `Ulcer` / `Infection` / `Normal` / `Gangrene`

Supplementary Dataset (Recommended for Better Training)

DFUC 2021 Challenge Dataset

15,683 DFU patches with the same 4-class structure. Requires application (usually approved within days for academic use).

Detail	Value
Total images	15,683 (5,955 train / 5,734 test / 3,994 unlabeled)
Format	JPEG patches
Access	Application required at dfu-challenge.github.io

Task: Multi-class classification (4 classes)

How it will be used: Train EfficientNet on 4-class DFU classification using the Kaggle dataset, optionally augmented with DFUC 2021 for better generalization. New agent `dfu_agent/`. Output: ulcer type + severity description. Update `image_classifier.py` to recognize foot wound images.

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13. Colon Polyp Detection

Kvasir-SEG — Simula Research Lab

What it contains: 1,000 gastrointestinal polyp images with corresponding pixel-level segmentation masks from colonoscopy procedures. Part of the larger Kvasir dataset.

Detail	Value
Total images	1,000
Masks	1,000 binary segmentation masks
Format	JPEG images + JPEG masks
Resolution	332×487 to 1920×1072
Size	~44 MB
Access	Free, direct download
License	CC BY 4.0

Task: Binary segmentation (polyp region vs. background)

Supplementary: **CVC-ClinicDB** — 612 colonoscopy frames with masks. Combine with Kvasir-SEG for a total ~1,612 samples.

How it will be used: Train UNet or DeepLabV3+ for polyp segmentation. New agent `polyp_agent/`. Output: segmentation overlay image showing detected polyp region. Update `image_classifier.py` to recognize colonoscopy frame images.

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14. Expanded Dermatology — ISIC Multi-class

HAM10000 / Skin Cancer MNIST — Kaggle

What it contains: 10,015 dermoscopic images spanning 7 types of skin lesions, collected from two major dermatoscopy sources (HAM10000 — Human Against Machine with 10,000 training images). A significant expansion over the current binary skin lesion agent.

Detail	Value
Total images	10,015

Format	JPEG, 600×450
Size	~2.4 GB
Access	Free on Kaggle
License	CC BY-NC 4.0

Classes (7 lesion types):

Code	Full Name	Malignancy
`mel`	Melanoma	Malignant
`nv`	Melanocytic nevi	Benign
`bcc`	Basal cell carcinoma	Malignant
`akiec`	Actinic keratoses / Intraepithelial carcinoma	Pre-malignant
`bkl`	Benign keratosis-like lesions	Benign
`df`	Dermatofibroma	Benign
`vasc`	Vascular lesions	Benign

Class imbalance: `nv` dominates (~67%). Use weighted sampling or focal loss.

Task: Multi-class classification (7 classes) — upgrades the current binary skin lesion agent

How it will be used: The current `SKIN_LESION_AGENT` performs segmentation only. This dataset enables a classification upgrade — identify the lesion type in addition to segmenting it. Add a classification head alongside the existing segmentation model, or train a separate classifier and run both. Human validation mandatory.

Summary Table

#	Disease	Dataset	Images	Size	Task	Diseases / Conditions Detected	Access	Agent Status
1	COVID-19 CXR	Internal	—	—	Classification	COVID-19 Positive, Normal	—	Done
2	Skin Lesion	ISIC	—	—	Segmentation	Lesion region (binary mask)	—	Done
3	Brain Tumor	Kaggle (Nickparvar)	7,023	~700 MB	Classification	Glioma, Meningioma, Pituitary Tumor, No Tumor	Free	Placeholder
4	Tuberculosis	NIH CXR14 (Kaggle) + Shenzhen/Montgomery (Kaggle)	112k+	~42 GB + ~1 GB	Classification	Tuberculosis, Normal	Free	Planned
5	Pneumothorax	SIIM-ACR (Kaggle)	12,954	~10 GB	Segmentation	Pneumothorax region, No Finding	Free	Planned
6	Multi-label CXR	CheXpert (Stanford)	224,316	~11 GB (small) / ~439 GB (full)	Multi-label Classification	Cardiomegaly, Edema, Consolidation, Atelectasis, Pleural Effusion + 9 more	Registration	Planned
7	Diabetic Retinopathy	APTOS 2019 (Kaggle)	3,662	~9 GB	Ordinal Classification	No DR, Mild DR, Moderate DR, Severe DR, Proliferative DR	Free	Planned
8	Mammography	CBIS-DDSM (Kaggle)	~10,239	~6.8 GB	Classification	Benign Mass, Malignant Mass,	Free	Planned

						Benign Calcification, Malignant Calcification		
9	Histopathology	Kather NCT-CRC (Zenodo)	100,000	~7.4 GB	Multi-class Classification	Adipose, Background, Debris, Lymphocytes, Mucus, Smooth Muscle, Normal Mucosa, Cancer Stroma, Tumor Epithelium	Free	Planned
10	Fracture	MURA (Stanford)	40,561	~6 GB	Classification	Normal, Abnormal — across Elbow, Finger, Forearm, Hand, Humerus, Shoulder, Wrist	Registration	Planned
11	AMD / OCT	Kermany 2018 (Kaggle)	84,495	~6.3 GB	Multi-class Classification	CNV (Wet AMD), DME (Diabetic Macular Edema), DRUSEN (Dry AMD), Normal	Free	Planned
12	Diabetic Foot Ulcer	laithjj (Kaggle) + DFUC 2021	1,426 + 15,683	~500 MB + ~2 GB	Classification	Ulcer, Infection, Normal, Gangrene	Free + Application	Planned
13	Colon Polyp	Kvasir-SEG + CVC-ClinicDB	1,000 + 612	~44 MB + ~100 MB	Segmentation	Polyp region (binary mask), No Polyp	Free	Planned
14	Expanded Dermatology	HAM10000 (Kaggle)	10,015	~2.4 GB	Multi-class Classification	Melanoma, Melanocytic Nevi, Basal Cell Carcinoma, Actinic Keratoses, Benign Keratosis, Dermatofibroma, Vascular Lesions	Free	Planned